



FACULTY OF COMPUTER SCIENCE AND MATHEMATICS

BACHELOR OF COMPUTER SCIENCE (SOFTWARE ENGINEERING) WITH
HONORS

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GROUP 7 (K2)

Cloud-Native Learning Management System (LMS)
(Cloud-Native / Layered Architecture)

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1.0 Selected Architecture Style

The Learning Management System (LMS) is designed using a three-tier layered architecture. In this architecture, the system is divided into three distinct layers, which are the presentation layer, the business layer, and the data access layer. In the first tier, the presentation layer is for managing user interaction and the user interface (UI). Meanwhile, the business layer is for implementing core functionalities, workflows, and rules. Lastly, the data access layer is responsible for storing and retrieving application data by using the Aiven cloud. Aiven makes our data access layer more reliable. It automatically scales and fixes itself if there are errors. Because it works on many cloud platforms, it makes the system more flexible and resilient.

By using Cloud-Native implementation, each layer is physically distributed and scaled independently using cloud services. The LMS is no longer a static application. Instead, it becomes a dynamic system of services. Using Aiven for the data tier provides a strong storage solution that supports the distributed nature of the learning system.

2.0 Problem Description

Learning Management Systems (LMS) are essential in supporting digital education by managing users, courses, learning materials, and assessments. However, many existing LMS platforms still face several challenges that affect their effectiveness. One major issue is the management of different user roles, such as students, lecturers, and administrators, which requires proper access control to avoid security risks. In addition, LMS platforms must handle large volumes of data, including course content and user activities, making it difficult to ensure data accuracy and real-time updates. Communication features such as messaging and file sharing also need to function smoothly, but integration between different system components can sometimes be inefficient.

Another significant challenge is system performance and scalability, especially during peak periods such as assignment submissions or online examinations. In such situations, many systems experience slow response times or system failures, which can disrupt the learning process. Furthermore, generating clear and meaningful reports from large amounts of data remains a challenge, affecting decision-making and overall system usability. These limitations highlight the

need for a more advanced solution using a cloud-native and layered architecture to improve system scalability, reliability, and overall user experience.

3.0 System Objectives

The objective of the Cloud-Native Learning Management System (LMS) is to provide high availability for the learning environment. By using a three-tier architecture, the system will achieve a total separation of concerns. This way, it will ensure that updates to the user interface do not interfere with business logic and data management.

Also, independent scalability is the key objective of this system. The system is designed to automatically assign resources to specific layers, such as the business layer, during peak usage periods. Finally, by integrating Aiven, the system aims to remain cloud-independent, that allow to operate across multiple cloud providers while maintaining automated data backups and self-healing capabilities.

4.0 Initial Functional Requirements

The learning management system prototype focuses on two core modules that demonstrate the full architectural flow from the user interface to the Aiven.io cloud database.

Module 1: User Authentication

- FR-01: User Registration

The system shall allow new users to register by providing a username, email and password, which are then securely stored in the Aiven.io, MYSQL database.

- FR-02: Secure Login Authentication

The system shall verify user credentials against the stored records in the cloud database to grant access to the system.

- FR-03: Session Initialization

Upon successful login, the system shall create a session object to store user identity and maintain the login state during navigation.

- FR-04: Secure Logout

The system shall provide a logout function that terminates the user's session and redirects the user back to the login page.

Module 2: Course Dashboard

- FR-05: Authenticated Dashboard Access

The system shall ensure that only users with an active session can access the Course Dashboard module.

- FR-06: Dynamic Course Retrieval

The system shall fetch and display a list of available courses from the Aiven.io database based on the user's profile or enrollment status.

- FR-07: Course Detail View

The system shall allow users to select a specific course to view detailed information, such as course objectives and descriptions.

- FR-08: Session-Data Integration

The system shall display personalized information on the dashboard by retrieving data from the active session created in Module 1.

5.0 Team Member Roles

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